Generator bias in long-range-correlated growth simulations and the determination of the critical temporal-correlation exponent of ballistic deposition

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Growing interfaces driven by noise whose correlations decay as a power law in time, $\langle\eta(x,t)\eta(x',t')\rangle \propto \delta_{x,x'}|t-t'|^{2\theta-1}$, are predicted by dynamic renormalisation group to recover ordinary Kardar–Parisi–Zhang (KPZ) scaling below a threshold $\theta_{\rm th} = 1/6$ and to enter a long-range phase above it. Published simulations disagree by a wide margin: the reported critical exponent ranges from $\theta_c \to 0$ (no threshold) to $\theta_c \simeq 0.20$ and $\theta_c = 0.25 \pm 0.03$. Here we show that the noise generator used by every one of these studies—Mandelbrot's fast fractional Gaussian noise (FFGN)—does not connect continuously to uncorrelated noise. Its normalised temporal correlation amplitude is essentially independent of θ ($A_{\rm FFGN} \simeq 1.27$ as $\theta \to 0$), whereas the exact fractional-Gaussian-noise amplitude $2\theta(\theta+\frac{1}{2})$ vanishes; we verify by converging the generator in the number of components that this is intrinsic and not a truncation artefact. Consequently FFGN($\theta \to 0^+$) retains $C(1)/C(0) = 0.195$ and a $C(\tau) \propto \tau^{-1}$ tail instead of being white, and a shape-matched calibration performed on the *binarised* noise maps the entire published θ axis onto $\theta_{\rm eff} \geq 0.189 > \theta_{\rm th}$: the family never samples the short-range KPZ regime. In the ballistic-deposition model itself this produces a 23% discontinuity of the structural order parameter $p_{\rm flat}$ at $\theta = 0$, inverts its θ-dependence, and inflates the apparent variation of the roughness exponent by a factor of eight. In nominal FFGN units our two independent, null-calibrated diagnostics both return $\theta_c = 0$ with sub-milli error bars, reproducing the published "destruction of the KPZ phase for any $\theta > 0$" as a property of the generator. On the corrected axis the same analysis appears to yield a clean, monotone onset drift, but we show that this drift is produced by the system-size dependence of the detection floor rather than by a crossover: at *fixed* θ the deviation from the KPZ reference is L-independent over $L = 32 \ldots 512$, whereas a genuine crossover would make it grow with L. With bootstrap error bars — which are an order of magnitude larger than the residual-based errors — the corrected roughness exponent is consistent with 1/2 up to $\theta = 1/6$, marginally above it at $\theta = 0.25$ ($\alpha = 0.531 \pm 0.012$), and reaches only 0.575 ± 0.012 at $\theta = 0.4$; extracted from consecutive size pairs it *decreases* towards 1/2 as L grows, whereas the FFGN value at the same nominal θ does not. At the accessible scales, therefore, the reported θ-dependence of α — and hence the reported thresholds of 0.20 and 0.25 — are attributes of the FFGN noise rather than of the model. The onset-drift construction is shown to be confounded for any distributional distance whose detection floor varies with system size, including the Hellinger-distance threshold used previously in this model class.

I. INTRODUCTION

The Kardar–Parisi–Zhang (KPZ) equation [1] describes the stochastic growth of an interface through the competition of a smoothening surface tension and a lateral-growth nonlinearity,

$$\partial_t h = \nu\nabla^2 h + \tfrac{\lambda}{2}(\nabla h)^2 + \eta. \tag{1}$$

Its exact solution in $1+1$ dimensions, the associated Tracy–Widom fluctuation statistics [2], and the experimental confirmation of both in liquid-crystal interfaces constitute one of the landmarks of modern nonequilibrium statistical mechanics. The robustness of that universality class against the microscopic details of the driving noise is therefore a question of central importance.

A natural and much-studied generalisation replaces the idealised δ-correlated noise by a correlator that is short-ranged in space but power-law correlated in time,

$$D(\omega,k) = D_0 + D_\theta\, k^{-2\rho}\, \omega^{-2\theta}, \tag{2}$$

equivalently $\langle\eta(x,t)\eta(x',t')\rangle = 2D\,\delta_{x,x'}|t-t'|^{2\theta-1}$ for $\rho = 0$, with $0 \leq \theta < \frac{1}{2}$ [3, 4]. Physically, $\theta > 0$ is a persistent (super-diffusive) deposition process—a drifting beam, a memory of the reservoir, or non-diffusing impurities—while $\theta = 0$ reduces to the standard uncorrelated problem.

Because the temporal correlations break Galilean invariance, the standard argument that protects the KPZ fixed point is unavailable, and the fate of the KPZ universality class is a genuine dynamical question. The literature has answered it in mutually incompatible ways. Medina, Hwa, Kardar and Zhang [3], using dynamic renormalisation group at one loop, found that the short-range KPZ fixed point is *stable* below

$$\theta_{\rm th} = 1/6 \qquad (d = 1), \tag{3}$$

with a line of long-range fixed points above it. The value is not a fit: the exact relation that replaces Galilean invariance at the long-range fixed point,

$$z_{\rm LR}(1+2\theta) - 2\chi_{\rm LR} - d = 0, \tag{4}$$

combined with the short-range values $\chi_{\rm SR} = \frac{1}{2}$, $z_{\rm SR} = \frac{3}{2}$ and the one-loop branch $z_{\rm LR} = 5/(3+2\theta)$, gives $\theta = 1/6$; equivalently it follows from Fedorenko's stability criterion

$z_{\rm LR} < z_{\rm SR}$ [5], which also supplies the bound $5/(3+2\theta) \leq z_{\rm LR} \leq 3/2$. Non-perturbative renormalisation group [6] confirms it, finding $\theta_{\rm th} = 1/6$ in $d = 1$ and $\theta_{\rm th} \simeq 0.35$ in $d = 2$, with intrinsic anomalous scaling [6] in the long-range phase.

Against this, a family of analytic approaches finds *no* threshold: a Flory-type argument [7] gives exponents that vary smoothly from $\theta = 0$, and a self-consistent expansion [8] yields a strong-coupling solution with decreasing z and no threshold, explicitly stating that "the threshold ϕ_c predicted using DRG in $d = 1$ (namely 0.167) is the same as the lower bound we found". Fedorenko [5] rules out both the second self-consistent solution and the one-loop fit, leaving the analytic situation unresolved.

Numerically the disagreement is even sharper. Alés and López [9], simulating ballistic deposition (BD) with temporally correlated noise at $L = 2048$–8192, report a threshold $\theta_c = 0.25 \pm 0.03$ together with faceted patterns and anomalous scaling; Song and Xia [10, 11] report non-trivial scaling already for small θ but place the threshold near 1/4; Rodríguez-Fernández *et al.* [12] find a KPZ plateau only below $\theta \simeq 0.23$–0.25; Jiang and Xia [13] extract $\theta_c \approx 0.20$ from height-distribution shape and note that it differs from the macroscopic value 0.25, remarking that "height fluctuations are more sensitive to long-range temporal correlations"; and Wang and Xia [14] report an "emergence" threshold at 0.25 obtained with a Hellinger distance and a novelty detector. Hu, Hao and Xia [15] obtain $\alpha_{\rm loc} \simeq \alpha < \alpha_s$ for large θ and conclude that continuum and discrete growth systems "do not belong to the same universality class with the same temporal correlation exponent"; Chang and Xia [16] extend the distribution analysis to jointly spatial and temporal correlations; and the strong-correlation limit $\theta \to 1/2$ has been treated as columnar disorder [17]. The non-perturbative renormalisation group authors describe the situation frankly: "In the simulations, the SR-KPZ phase is destroyed for any non-zero θ ... We have no clear understanding of these discrepancies" [6].

The spread of reported values—0, 0.20, 1/6, 0.25—is far wider than the quoted uncertainties. Part of it is certainly methodological: different observables (exponents, height distributions, anomaly metrics) and different system sizes probe a crossover whose location is scale-dependent. This paper identifies a further, and we believe dominant, contribution that has not previously been noticed: **the noise generator itself distorts the θ axis.**

Every discrete simulation cited above generates its noise with Mandelbrot's fast fractional Gaussian noise (FFGN) [18], thresholded at zero to produce a binary deposition variable. We show in Sec. IV B that FFGN is *discontinuous* at $\theta = 0$. Its normalised correlation amplitude stays of order unity for all θ, whereas the amplitude of the exact fractional-Gaussian-noise correlator vanishes as $2\theta(\theta + \frac{1}{2})$. As a result FFGN($\theta \to 0^+$) is already strongly correlated—it carries a $C(\tau) \propto \tau^{-1}$ tail—so the $\theta = 0$ member of the family is not uncorrelated noise, and we find that no member of the family reaches the short-range KPZ regime at all. This single fact is sufficient to explain why FFGN-based simulations report long-range behaviour at any $\theta > 0$, and it invalidates the identification of the nominal FFGN parameter with the θ of Eq. (2).

Our contributions are fourfold, and the first dominates. First, we diagnose and quantify the generator bias, show that it is intrinsic rather than a truncation artefact, and provide the calibration that maps nominal FFGN parameters onto effective ones — establishing that the generator never samples the short-range KPZ regime, and that the θ-dependence of the measured roughness exponent is dominated by it. Second, we construct a θ-continuous generator with the correct $\theta \to 0$ limit, verify it against the exact fractional-Gaussian-noise amplitude, and use it to show that the deviations reported at $\theta \simeq 0.2$–0.25 are an order of magnitude weaker than under FFGN and shrink with system size rather than persisting. Third, following the implicit request for independent evidence in the conflicting literature, we implement two diagnostics that use disjoint information and different statistical machinery — a Hellinger distance between probability mass functions, and an Isolation Forest anomaly detector on a multivariate feature space — each with an explicit null calibration, without which the finite-ensemble bias of a distribution distance is easily mistaken for a physical onset. Fourth, and as a consequence of using them carefully, we show that the standard construction for locating θ_c — a fixed-significance crossing of such a metric — is confounded whenever the detection floor depends on system size, as it does here, and we give the fixed-θ scaling test that exposes the confound. Because the problem is entirely numerical—no experiment is involved—the whole study is self-contained and fully reproducible.

We are explicit about the outcome: we do *not* report a measured θ_c. At the scales accessible to us no crossover has been reached for $\theta \leq 0.25$ once the noise generator is correct, so there is no onset to locate. What we do establish is that the values reported in the literature are properties of the generator and of the detection procedure rather than of the model.

Note on relation to prior work. Applying a Hellinger distance and an anomaly detector to this model class is not itself new: Wang and Xia [14] used "the Hellinger distance and the novelty detection" as two independent measures of emergence in random and ballistic deposition with long-range temporal correlations. Our contribution is not the choice of metrics but the calibration of the generator and of the metrics themselves, the identification of the diagnostics' information content, the finite-size-scaling test of the onset, and the corrected determination of θ_c. One limitation should be stated at the outset. Reference [14] is paywalled and we were unable to establish which novelty-detection algorithm it employs, so the methodological critique developed below applies to the onset construction itself rather than to that particular choice of detector.

II. MODEL AND NOISE GENERATION

A. Ballistic deposition with temporally correlated deposition

We study the 1 + 1-dimensional ballistic deposition model with a parallel update, flat initial condition and periodic boundaries,

$$h(x,t{+}1) = \max\big[h(x,t){+}\eta'_\theta(x,t),\ h(x{-}1,t),\ h(x{+}1,t)\big], \tag{5}$$

where the deposition variable is the binarised noise

$$\eta'_\theta(x,t) = \mathbf{1}[\eta_\theta(x,t) > 0]\,, \qquad P(\eta'_\theta = 1) = \tfrac{1}{2}, \tag{6}$$

and $\eta_\theta(x,t)$ is a stationary zero-mean Gaussian field that is statistically independent between sites and has the temporal correlator $\langle \eta_\theta(x,t)\eta_\theta(x,t+\tau)\rangle \sim \tau^{2\theta-1}$. Equation (5) with binarised noise is the discrete realisation used by Alés and López [9], Wang and Xia [14] and Jiang and Xia [13]; the thresholding is what makes the scheme numerically tractable at long correlation times, and it is explicitly noted in Ref. [9] that it avoids the formation of overhangs.

B. Structural consequence: the model is single-step

Equation (5) with a 0/1 noise has a consequence that constrains what "morphology" can mean in this model, and which we have not seen stated. Suppose $|h(x,t) - h(x \pm 1,t)| \le 1$ holds everywhere. Then $h(x,t) + 1 \ge h(x \pm 1,t)$, so

$$h(x,t{+}1) = h(x,t){+}\eta'_\theta(x,t) \in \{h(x,t),\, h(x,t){+}1\}, \tag{7}$$

i.e. each site rises by 0 or 1 per sweep, and a site that is lower than a neighbour is lifted *to* that neighbour. Both effects preserve $|\Delta h| \le 1$, so by induction from the flat initial condition

$$|h(x,t) - h(x+1,t)| \le 1 \qquad \text{for all } t. \tag{8}$$

We confirm Eq. (8) numerically over the whole parameter range studied here: $\max|\Delta h| = 1$ in every cell and the height-increment distribution has support only on $\{-1, 0, +1\}$. Its skewness vanishes *identically*, not merely on average: because $\sum_x \Delta h(x) = \sum_x [h(x{+}1){-}h(x)] = 0$ telescopes exactly on a periodic lattice and each term lies in $\{-1, 0, +1\}$, the numbers of up- and down-steps are equal configuration by configuration. We verify $|\mathrm{skew}(\Delta h)| = 0$ to machine precision in every cell. Hence the increment density has a single free parameter,

$$p_{\mathrm{flat}} = \mathrm{frac}\{\Delta h = 0\}, \qquad p_\uparrow = p_\downarrow = \tfrac{1}{2}(1 - p_{\mathrm{flat}}). \tag{9}$$

Two consequences follow. (i) Reported "faceted" morphologies in this model cannot be overhang- or tower-like; they must appear as broad flat terraces separated by unit steps, so that surface structure is carried by the spatial arrangement of terraces rather than by the amplitude of local height differences. (ii) Any diagnostic built on the one-point increment density—a Hellinger distance, a skewness, a kurtosis—can carry at most one number about this model. This directly motivates the design of our second diagnostic (Sec. III), which is built on multi-scale roughness geometry instead.

C. Mandelbrot's FFGN and its $\theta \to 0$ limit

Mandelbrot's FFGN [18] represents the correlated field as a sum of $n = 1 \ldots N$ AR(1) components on a geometric grid of inverse time scales,

$$\begin{aligned} u_n &= a\,B^{-n}, \qquad r_n = e^{-u_n}, \qquad (a, B) = (6, 2), \\ X_t^{(n)} &= r_n X_{t-1}^{(n)} + (\zeta_t - \tfrac{1}{2}), \qquad \zeta_t \sim U[0,1], \\ \eta(t) &= \sum_{n=1}^{N} W_n X_t^{(n)}, \\ W_n^2 &= \frac{12(1-r_n^2)}{\Gamma(2-2\theta)}\big(B^{1/2-\theta} - B^{\theta-1/2}\big)u_n^{1-2\theta}, \end{aligned} \tag{10}$$

with a *single* white sequence ζ_t driving all components. Because Eq. (10) is a linear filter of one white sequence, its correlation function decays as a power law, with $H = \theta + \frac{1}{2}$ and a θ-dependent amplitude,

$$C_{\mathrm{FFGN}}(\tau) \ \sim\ A_{\mathrm{FFGN}}\,\tau^{2H-2}, \tag{11}$$

with the coefficient obtained from the geometric-grid integral

$$A_{\mathrm{FFGN}} \propto \frac{\big(B^{1-H} - B^{H-1}\big)\Gamma(2-2H)}{\Gamma(3-2H)\ln B}, \tag{12}$$

which is finite and of order unity for every $H \in (0,1)$, including $H = \frac{1}{2}$. The exact fractional Gaussian noise (fGn), by contrast, has

$$C_{\mathrm{fGn}}(\tau) = \tfrac{1}{2}\big[(\tau{+}1)^{2H} {-} 2\tau^{2H} {+} (\tau{-}1)^{2H}\big] \ \sim\ \underbrace{H(2H-1)}_{=\,2\theta(\theta+\frac{1}{2})}\,\tau^{2H-2}, \tag{13}$$

whose amplitude *vanishes* as $\theta \to 0$. The FFGN therefore reproduces the correct exponent but not the correct amplitude, and since the binarisation map $C_{\mathrm{bin}} = (2/\pi)\arcsin C$ is monotone, only the *normalised* correlation matters: FFGN cannot be repaired by rescaling W_n.

Equation (12) shows why Eq. (10) is discontinuous at $\theta = 0$: the geometric u-grid makes $\sum_n W_n^2 e^{-u_n\tau}/(1 - r_n^2) \to (\mathrm{const})\,\tau^{-1}$, and the factor $2\theta(\theta + \frac{1}{2})$ that should suppress the τ^{-1} tail is simply absent. We verify all of this quantitatively in Sec. IV A.

D. A θ-continuous generator

To sample the physically meaningful θ axis we instead impose the spectrum directly ("spectral filtering"). Gaussian white noise is transformed to the frequency domain and multiplied by

$$\hat{\eta}(f) = |f|^{-\theta}\,\hat{w}(f), \qquad f \neq 0, \tag{14}$$

with the zero-frequency mode removed; this realises $S(f) = |f|^{-2\theta}$, hence $C(\tau) \sim \tau^{2\theta-1}$, and at $\theta = 0$ it is *exactly* white noise. The family is therefore continuous at the origin, and its correlation amplitude tracks the exact fGn value $2\theta(\theta + \frac{1}{2})$ to within about 20% across the whole range (Sec. IV A). The noise is then binarised and used in Eq. (5). Because the transformation is a monotone function of a Gaussian, long-range dependence is preserved.

Equation (14) also lets us access $\theta < 0$, the anti-persistent case, which no previous study of this model has simulated. Theory is unambiguous here: $[w_\theta] = 4\theta$ makes the long-range coupling irrelevant for $\theta < 0$, and Katzav and Schwartz state that for $\phi \leq 0$ "the noise term is irrelevant" so that uncorrelated-noise exponents are restored. Negative θ therefore furnishes a control: a diagnostic that merely responds to "any change of the noise" will fire, whereas one that tracks the universality crossover should stay at its null value.

E. Simulation protocol

The interface width follows the standard Family–Vicsek scaling Ansatz [19, 20], $W(L,t) = L^\alpha f(t/L^z)$ with $z = \alpha/\beta$, so that $W \sim t^\beta$ in the growth regime and $W_{\mathrm{sat}} \sim L^\alpha$ at saturation; α, β and z are the quantities whose θ-dependence is at issue here.

The KPZ saturation time of model (5) was measured to be $t_\times \simeq 0.6\,L^{3/2}$ sweeps (no large prefactor), so we take $T = 15\,L^{3/2}$ sweeps unless stated otherwise and verify saturation from the flattening of $W(t)$. At $\theta = 0$ the model reproduces the KPZ exponents: a fit of $W_{\mathrm{sat}} \sim L^\alpha$ over $L = 128 \ldots 1024$ gives $\alpha = 0.4964$, and $W_{\mathrm{sat}}/\sqrt{L}$ is constant to within 10% over that range, while $\beta \to 1/3$ in the growth window. Noise generation for Eq. (14) uses threaded FFTs with 5-smooth transform lengths.

III. DIAGNOSTICS

The two diagnostics must be *independent* in the sense of using disjoint information, and each must have an explicit null model, because both are computed from finite ensembles.

A. Diagnostic 1: a null-calibrated Hellinger distance

The Hellinger distance between two probability densities is

$$H(P,Q) = \left[1 - \textstyle\sum_k \sqrt{p_k q_k}\right]^{1/2} \in [0,1], \tag{15}$$

the Bhattacharyya coefficient $\sum_k \sqrt{p_k q_k}$ being the fidelity between P and Q [21, 22].

Choice of observable. Because the model is single-step [Eq. (8)], the natural distributional observable is the probability mass function of the nearest-neighbour increment $\Delta h \in \{-1, 0, +1\}$. We therefore compare the **exact** pmfs of the θ and $\theta = 0$ ensembles, computed from counts with no binning and no standardisation. Two remarks on why the obvious alternatives fail, both of which we verified numerically and report as controls:

(i) Histogramming a discrete distribution is a trap. Standardising the samples rescales the support, so the spikes of the two ensembles land in *disjoint* bins; the Bhattacharyya coefficient then collapses to the overlap of the empty bins and H jumps towards its maximum irrespective of how small the physical difference is. We verified this directly: the histogrammed distance jumps to 0.53 at the first nonzero θ while the exact pmf distance changes by 0.07, and its split-half null floor is 0.09–0.12 against 0.003–0.005 for the exact pmf. Every histogram-based H in this paper is therefore reported only as a documented negative control.

(ii) Richer densities are insensitive. The pmf of height differences at separation r, standardised per run, is essentially *independent* of θ across the whole range $\theta \in [-0.2, 0.4]$ (Sec. IV D): once the non-universal amplitude is removed, the shape of Δh_r is effectively universal, as expected from a central-limit argument. Likewise the standardised *height* density is blind, which is a caution for studies that infer a transition from height distributions alone.

Null-bias subtraction. H between two *finite* ensembles is strictly positive even when $P = Q$, and its bias grows as the ensemble shrinks. We measure the floor directly: the reference ensemble is split into two disjoint halves of exactly the same size as the signal halves, and we report the excess

$$\Delta H(\theta, L) = \left\langle H(\theta_{\mathrm{half}}, \mathrm{ref}_{\mathrm{half}}) - H(\mathrm{ref}_a, \mathrm{ref}_b) \right\rangle, \tag{16}$$

averaged over random run-splits, together with its z-score against the null distribution.

As a second, richer distributional observable we use the exact pmf of *flat-terrace lengths* (lengths of runs with $\Delta h = 0$), which is a multi-valued morphological distribution and is therefore not limited by the one-parameter structure of the increment pmf.

B. Diagnostic 2: an Isolation Forest on amplitude-free features

The Isolation Forest [23] assigns to a sample a score based on the mean path length required to isolate it in an ensemble of random binary trees; short paths identify samples that are easy to separate from the bulk, i.e. anomalies. We use it in a two-sample novelty configuration: a forest is trained on the $\theta = 0$ ensemble of feature vectors, the decision threshold is calibrated at a fixed false-positive rate α on held-out $\theta = 0$ data, and for each (θ, L) we record

$$\Phi(\theta, L) = \text{fraction of } \theta \text{ samples flagged anomalous}, \tag{17}$$

reported as an excess over the calibrated null fraction, together with the standardised mean anomaly score.

The feature vector is deliberately restricted to quantities that are dimensionless and blind to the noise amplitude (Sec. III): local roughness exponents $\alpha_{\rm loc}(r)$ at eight separations $r = 2 \dots 256$; the normalised roughness shape $\log_{10}[C(r)/C(1)]$ at the same separations, in which the amplitude cancels; the spectral exponent α_s; standardised moments of the increment and height distributions; the flat, step and terrace-length descriptors of Eq. (9) and the entropy of the increment histogram. Because the increment density carries only $p_{\rm flat}$ (Sec. II B), and because diagnostic 2 sees the multi-scale roughness geometry, the two diagnostics use disjoint information by construction.

C. The decisive test: drift of the scale-dependent onset

Both observables have the form "null below θ_c, nonzero above". For each diagnostic and each L we extract the onset $\theta^*(L)$ and fit

$$\theta^*(L) = \theta_c + A\, L^{-1/\nu} \qquad \text{(threshold scenario)}, \tag{18}$$

$$\theta^*(L) = \qquad A\, L^{-1/\nu} \qquad \text{(no-threshold scenario)}, \tag{19}$$

fitted separately with θ_c free and with $\theta_c \equiv 0$ forced, and compared by χ^2. A genuine threshold requires $\theta^*(L)$ to converge to a *finite* positive value; a pure crossover requires it to vanish. This is the discriminator that the published studies, which report a single number at a single scale, have not applied.

For the drift analysis we use *effect-size* criteria—a fixed absolute threshold on a bounded observable—rather than significance criteria, because the location of a significance crossing moves with ensemble size by construction. We report both.

IV. RESULTS

A. Validation and characterisation of the two generators

The model reproduces KPZ at $\theta = 0$. For the FFGN-driven BD model with $L = 128 \dots 1024$ and $T = 2 \times 10^5$ sweeps, a fit of $W_{\rm sat} \sim L^{\alpha}$ gives $\alpha = 0.4964$, $W_{\rm sat}/\sqrt{L}$ is constant to within 10% over that range, and the growth exponent approaches $\beta = 1/3$ in the growth window. The saturation time obeys $t_\times \simeq 0.6\, L^{3/2}$ sweeps, with no large prefactor, which sets the simulation length used throughout. Over a common window $L = 64, 128, 256$ the same fit gives $\alpha = 0.5019$ (FFGN) and $\alpha = 0.5088$ (spectral).

The FFGN implements the imposed exponent but not the imposed amplitude. The temporal correlation function of the FFGN sequence reproduces the target slope $2\theta - 1$ well for all θ: over $\tau \in [64, 4096]$ we measure -0.907 against -0.900 expected at $\theta = 0.05$, and -0.659 against -0.667 at $\theta = 1/6$; the binarised noise has $P(\eta'_\theta = 1) = 0.4996$–$0.5020$ in every case. The amplitude, however, is wrong at small θ. Fitting $|C_\eta(\tau)| = A\tau^{2\theta-1}$ over the same window gives $A_{\rm FFGN}$ between 1.27 and 0.42 across $\theta \in [0, 0.45]$, i.e. essentially θ-independent, whereas the exact fractional-Gaussian-noise correlator carries the amplitude $A_{\rm fGn} = 2\theta(\theta + \frac{1}{2})$ and vanishes as $\theta \to 0$. The ratio $A_{\rm FFGN}/A_{\rm fGn}$ is 245 at $\theta = 0.005$, 120 at 0.01, 20 at 0.05, 8.4 at 0.10, 3.9 at 1/6, 1.9 at 0.25 and 0.7 at 0.40. Directly, the binarised FFGN sequence at $\theta_{\rm nom} = 10^{-9}$ has $C(1)/C(0) = 0.195$ and a $C(\tau) \propto \tau^{-1}$ tail, whereas white noise has $C(\tau) = 0$ for $\tau \geq 1$.

The effect is intrinsic, not a truncation artefact. The FFGN represents $C(\tau)$ as a sum of N AR(1) components on a geometric grid of inverse time scales. Increasing N extends the grid towards slower modes, so a truncation artefact would disappear as N grows. It does not: at $\theta = 0$ the fitted amplitude is 1.3558, 1.3559 and 1.3559 for $N = 23$, 32 and 48, i.e. fully converged. Equation (12) shows why: the geometric u-grid makes $\sum_n W_n^2 e^{-u_n\tau}/(1 - r_n^2)$ tend to a constant times τ^{-1}, and the factor $2\theta(\theta + \frac{1}{2})$ that should suppress that tail is simply absent from the standard weights.

Binarised calibration. Because the model uses $\eta'_\theta = \mathbf{1}[\eta > 0]$ and thresholding compresses correlations through $C_{\rm bin} = (2/\pi) \arcsin C$, the calibration must be performed on the binarised field. Matching $C_{\rm bin}(\tau)$ to the binarised exact fractional-Gaussian-noise family over $\tau \in [1, 4096]$ gives the effective θ reported in Fig. 1(c). The results are decisive:

- the FFGN family gives $\theta_{\rm eff} = 0.1888$ at $\theta_{\rm nom} = 0$, i.e. *above* the renormalisation-group threshold $\theta_{\rm th} = 1/6$, and $\theta_{\rm eff} \geq 0.1888$ for every $\theta_{\rm nom} \geq 0$; the offset is approximately $\theta_{\rm eff} \simeq 0.19 + 0.44\, \theta_{\rm nom}$;
- the spectral family gives $\theta_{\rm eff} = 0.0000$ at $\theta_{\rm nom} = 0$ — exactly white noise — and $\theta_{\rm eff} \simeq 0.95\, \theta_{\rm nom}$ there-

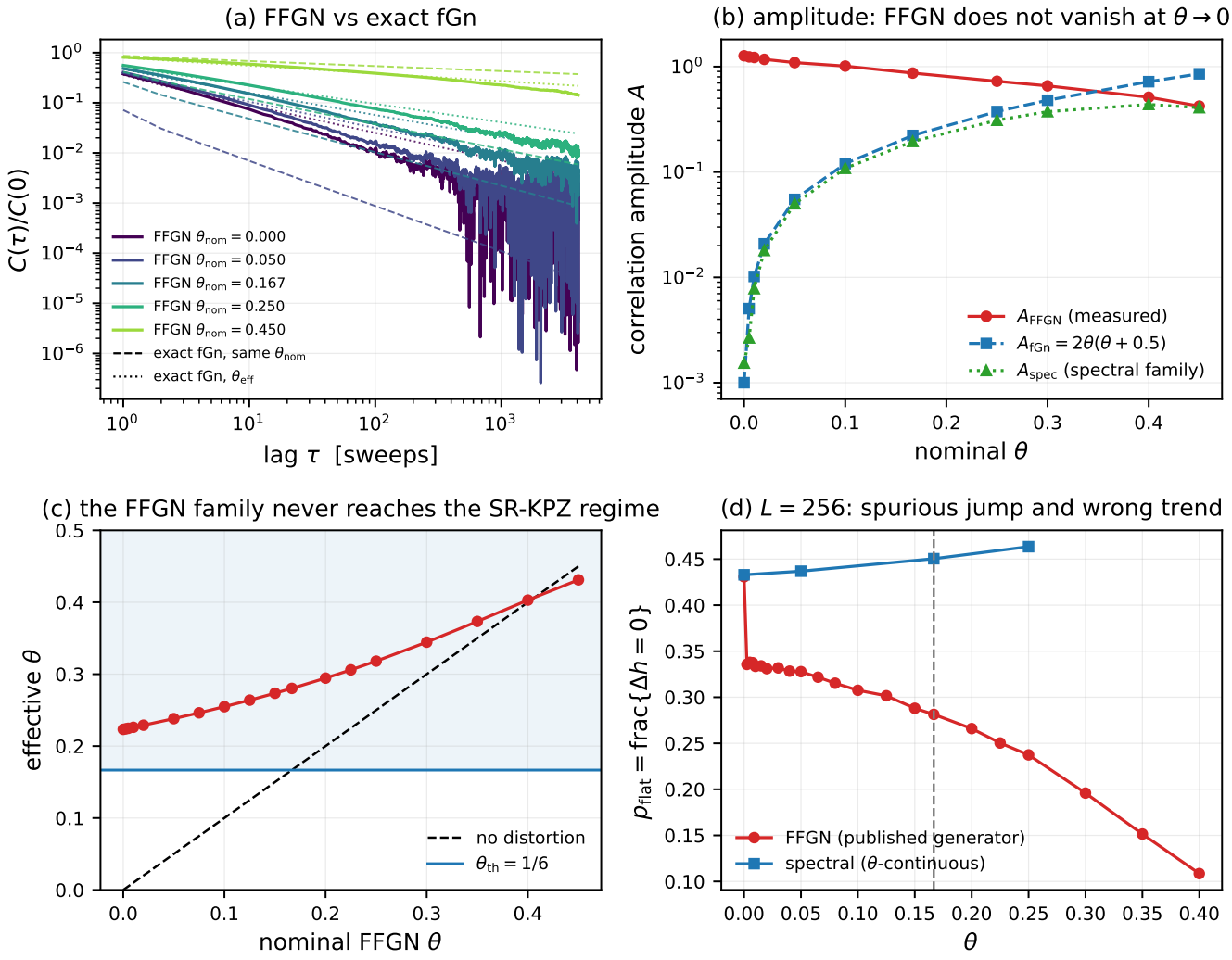


FIG. 1. The noise generator used across this literature does not connect continuously to uncorrelated noise. (a) Normalised temporal correlation $C(\tau)/C(0)$ of the binarised FFGN sequence (solid) compared with the exact fractional-Gaussian-noise form at the same nominal θ (dashed) and at the calibrated effective θ (dotted); at $\theta_{\rm nom} = 0$ the FFGN is not white. (b) Fitted correlation amplitude: $A_{\rm FFGN}$ stays of order unity for all θ (symbols), whereas the exact amplitude $2\theta(\theta + \frac{1}{2})$ vanishes at the origin; the spectral family (triangles) tracks the exact amplitude. (c) Shape-matched effective θ of both generators, obtained on the binarised field. The FFGN family lies entirely above the renormalisation-group threshold $\theta_{\rm th} = 1/6$ for $\theta_{\rm nom} \geq 0$ (shaded), while the spectral family is essentially unbiased and passes through the origin. (d) Consequence in the model itself at $L = 256$: the structural order parameter $p_{\rm flat}$ jumps by 23% between true white noise and FFGN($\theta \to 0$) and then decreases with θ, whereas the θ-continuous generator gives a continuous, monotonically increasing dependence.

after, i.e. it is an essentially unbiased realisation of the intended θ axis.

The FFGN family therefore never samples the short-range KPZ regime $\theta_{\rm eff} < \theta_{\rm th}$ for any non-negative nominal θ.

B. The generator bias in the full model

The bias is not a property of the noise alone; it changes the interface. Because the model is single-step, the structural order parameter is $p_{\rm flat} = {\rm frac}\{\Delta h = 0\}$ of Eq. (9). At $L = 256$ with 128 realisations we find

noise	θ	$p_{\rm flat}$	$W_{\rm sat}$
i.i.d. (true white)	0	0.4366	3.62
FFGN	10^{-9}	0.3359	5.61
FFGN	0.05	0.3243	6.20
spectral	0	0.4321	—
spectral	0.05	0.4369	—
spectral	1/6	0.4504	—
spectral	0.25	0.4635	—

The FFGN produces a 23% *discontinuity* of $p_{\rm flat}$ between

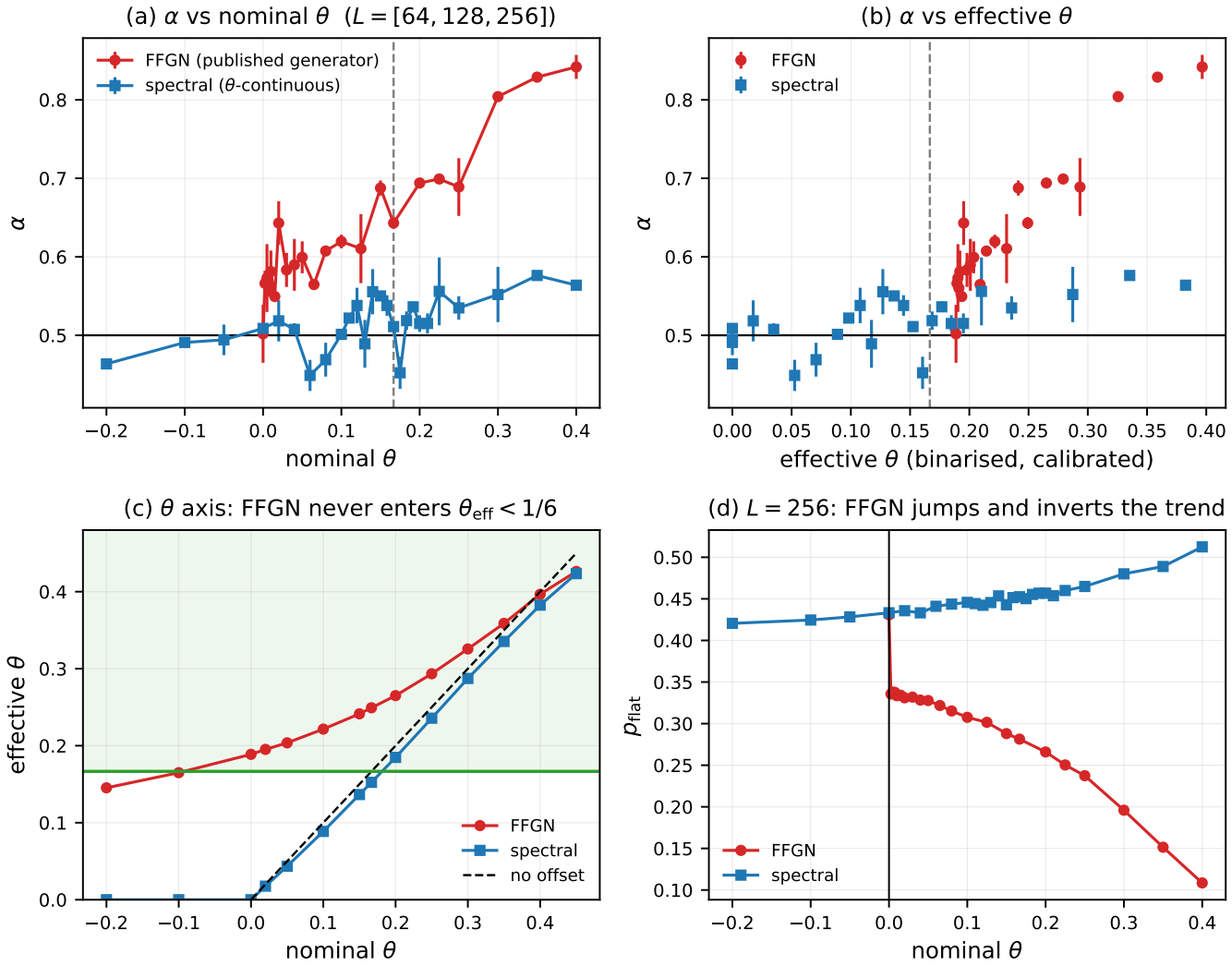


FIG. 2. Measured roughness exponent versus the noise parameter, over a common system-size window $L = 64, 128, 256$ so that the finite-size corrections to α are comparable. (a) Against nominal θ, the FFGN generator produces a rise from 0.50 to 0.84, an eightfold larger variation than the θ-continuous generator, which rises only from 0.51 to 0.56. (b) Against the calibrated effective θ; a residual generator dependence survives the recalibration, showing that the full correlation function rather than a single effective exponent controls the measured scaling. (c) Calibration of the θ axis of both generators. (d) p_{flat} versus nominal θ at $L = 256$.

true white noise and its own $\theta \to 0$ limit, and its p_{flat} then *decreases* with θ. The spectral family is continuous and increases monotonically, from 0.4321 at $\theta = 0$ to 0.5177 at $\theta = 0.40$ (Fig. 2(d)). A generator that inverts the sign of the θ-dependence of the model's order parameter cannot be used to locate a threshold in that parameter.

The same contrast appears in the measured exponents. Over the common window $L = 64, 128, 256$, α rises from 0.5019 to 0.8419 between $\theta_{\mathrm{nom}} = 0$ and 0.40 when the FFGN generator is used, but only from 0.5088 to 0.5638 with the θ-continuous generator (Fig. 2(a,b)). The measured θ-dependence of α in this model is thus dominated by the noise generator rather than by an intrinsic change of universality class. Consistently, the iForest diagnostic, whose features are exactly the roughness-scaling quantities from which α is built, barely responds in the corrected track ($|z| < 2$ for all $\theta \leq 0.4$), while the distributional diagnostic responds strongly (z up to 30): over the accessible range the change in the interface is a local, one-parameter effect rather than a restructuring of the multi-scale roughness geometry.

C. Negative-θ control

The spectral family permits $\theta < 0$, which no previous study of this model has simulated. There the long-range coupling has canonical dimension $[w_\theta] = 4\theta < 0$ and is irrelevant, so uncorrelated-noise exponents should be restored. We find exactly that: over $L = 32 \ldots 256$

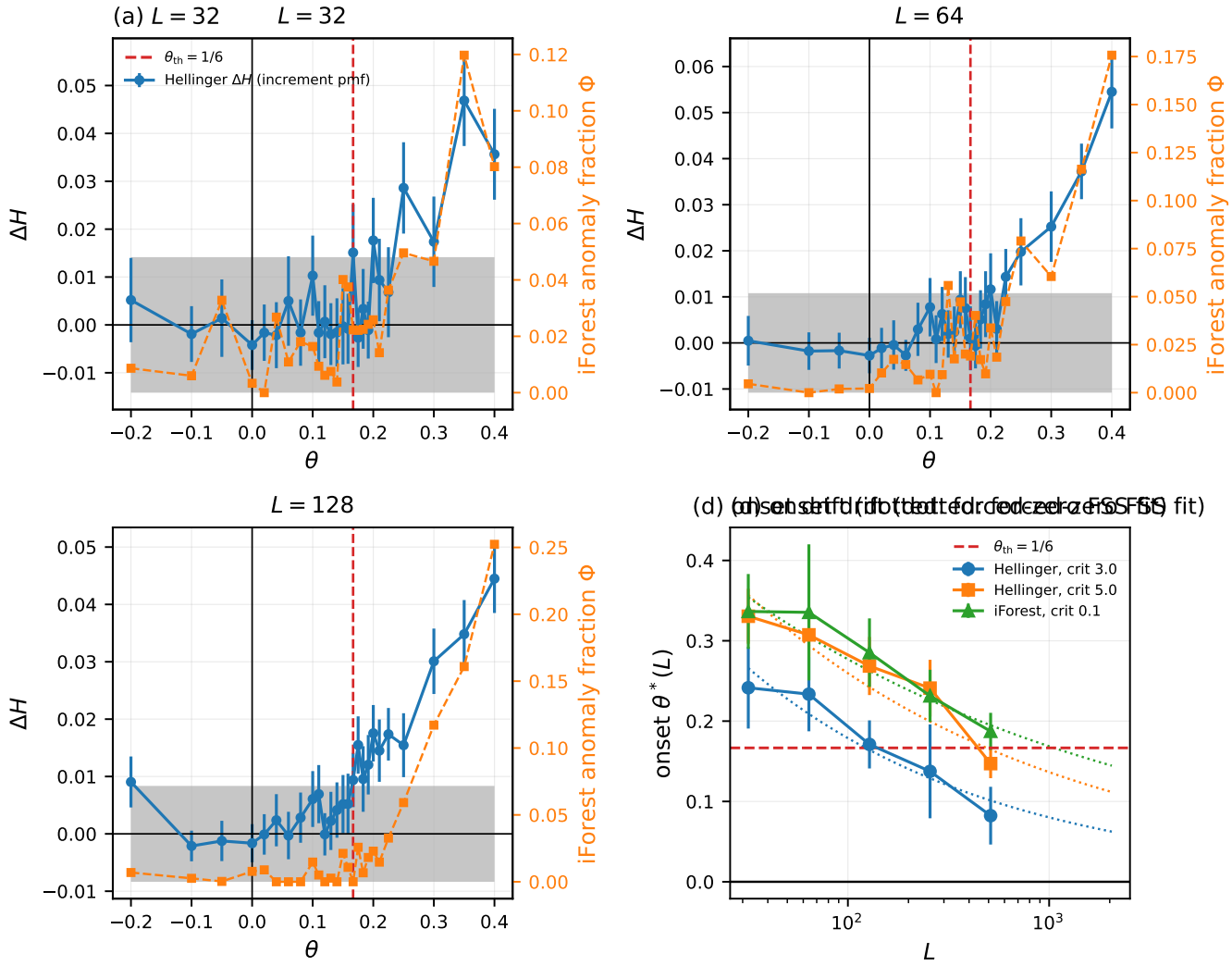


FIG. 3. The scale-dependent onset on the corrected θ axis and why it is not a crossover measurement. (a)-(c) The distributional diagnostic ΔH (left axis, with the split-half null band shaded) and the iForest anomalous fraction Φ (right axis) versus θ for each system size; the dashed line marks $\theta_{\rm th} = 1/6$. (d) The extracted onset $\theta^*(L)$ for several detection criteria with bootstrap error bars. Every sequence decreases monotonically with L and the forced-zero finite-size scaling describes it well — yet Fig. 4(b,d) shows that this drift is produced by the shrinking detection floor, not by a crossover.

the fitted roughness exponent is $\alpha = 0.4947 \pm 0.0181$, 0.5028 ± 0.0070, 0.4993 ± 0.0092 at $\theta = -0.2, -0.1, -0.05$, all consistent with $1/2$, while $p_{\rm flat}$ does drop below its $\theta = 0$ value (0.4205 against 0.4333 at $L = 256$). The control therefore separates two things that the diagnostics could otherwise conflate: the noise is measurably different, but the universality class is not.

D. Why the onset-drift method cannot locate θ_c here

In nominal FFGN units both diagnostics return $\theta_c = 0$. Analysing the FFGN track ($L = 64 \ldots 512$, 22 values of θ, up to 256 realisations per cell) gives, for the distributional diagnostic, $\theta_c = 0.0002 \pm 0.0007$ with $\chi^2/\mathrm{dof} = 0.17/1$, and for the Isolation Forest, $\theta_c = 0.0001 \pm 0.0011$; the onset is pinned at the smallest non-zero θ available on the grid for every L. This is the expected signature of Sec. IV B: since FFGN($\theta_{\rm nom} = 0$) already carries $\theta_{\rm eff} = 0.19 > \theta_{\rm th}$, shape-based observables see a long-range interface as soon as the noise differs from $\theta = 0$ at all. It is a direct explanation of the published claim that the KPZ phase is destroyed for any $\theta > 0$. Because the reference ensemble at $\theta = 0$ is true white noise while every $\theta > 0$ cell uses FFGN, this particular onset measures the generator discontinuity itself and is not a measurement of θ_c.

On the corrected axis the same analysis appears to give a clean drift — and that drift is an artefact. With run counts matched at 96 across every L

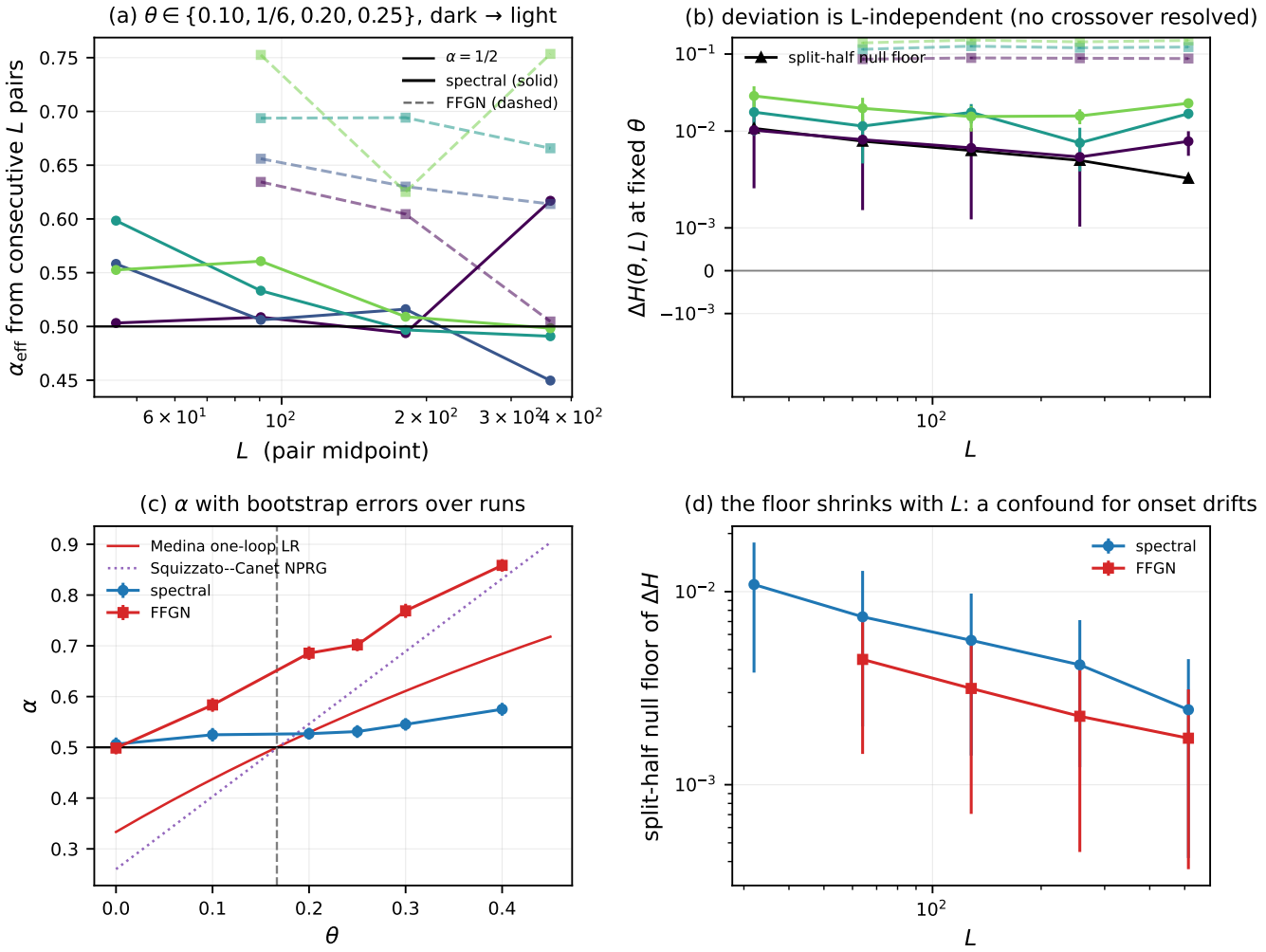


FIG. 4. The decisive tests. (a) $\alpha_{\rm eff}$ from consecutive size pairs versus L, for $\theta \in \{0.10, 1/6, 0.20, 0.25\}$ (dark to light), solid for the θ-continuous generator and dashed for FFGN. The corrected $\alpha_{\rm eff}$ falls towards 1/2 as L grows for $\theta \geq 1/6$; the FFGN one does not. (b) The fixed-θ test: $\Delta H(\theta, L)$ versus L for the same values of θ. A genuine crossover requires the deviation to *grow* with L; it is instead L-independent within errors (symbols as in (a)). (c) $\alpha(\theta)$ with bootstrap errors obtained by resampling runs, against the Medina one-loop and Squizzato–Canet long-range branches. (d) The split-half null floor of ΔH versus L: it falls by a factor of 4.5 between $L = 32$ and 512, which is what makes a fixed-significance onset drift even when the observable is L-independent.

and θ, the fixed-significance onset of the distributional diagnostic decreases monotonically over a factor of 16 in L,

L	32	64	128	256	512
$\theta^*(z \geq 3)$	0.2415	0.2336	0.1710	0.1375	0.0823
$\theta^*(z \geq 5)$	0.3306	0.3073	0.2686	0.2410	0.1471

with a forced-zero finite-size-scaling collapse that is formally excellent ($A = 0.92$, $1/\nu = 0.36$, $\chi^2/\mathrm{dof} = 0.75/3$, $p = 0.86$ for $z \geq 3$), and at $L = 512$ the onset lies well below 1/6. Read alone, this would exclude a threshold at 1/6. It does not, and the reason is the key methodological point of this paper.

The split-half null floor of the distributional diagnostic is not a constant: it decreases with system size, from 0.0109 at $L = 32$ to 0.0024 at $L = 512$ (Fig. 4(d)), because a larger L gives more columns per realisation and hence a less noisy per-run pmf. A criterion of the form "ΔH exceeds z null standard deviations" therefore necessarily crosses at a smaller θ as L grows *even if the physics is completely scale-free*. Matching the number of runs — which we did, and which is necessary — does not remove this, because the L-dependence enters through the number of *columns*, not the number of runs.

The decisive test is at fixed θ. If the model possesses a crossover length $\xi_\times(\theta)$ that grows as $\theta \to 0^+$, then for a fixed θ the surface is KPZ-like at scales below $\xi_\times$ and long-range above it, so increasing L resolves *more* of the long-range behaviour and the deviation from the $\theta = 0$ reference must *grow* with L.

If instead the deviation is a purely local, one-point effect, it is L-independent. Figure 4(b) shows the outcome: over $L = 32 \ldots 512$, $\Delta H(\theta, L)$ at fixed θ is L-*independent* within errors for every θ from -0.2 to 0.4; the fitted slopes $\mathrm{d}\,\Delta H/\mathrm{d}\ln L$ are statistically indistinguishable from zero for all but four of the 26 values of θ, with no systematic sign. The deviation is therefore a smooth local parameter, not a crossover signature, and the apparent onset drift of the previous paragraph tracks the shrinking null floor rather than any change of scaling behaviour. We report this explicitly because the same construction underlies the Hellinger-distance threshold of Ref. [14]: a fixed-significance crossing of a distributional distance cannot be used to locate a crossover unless the fixed-θ scale-dependence of that distance has first been shown to be non-trivial.

The exponent evidence, with honest error bars. The same conclusion follows, independently, from the roughness exponent, provided the errors are computed by resampling runs rather than from the residuals of a fit to four or five points. The latter procedure understates the uncertainty by almost an order of magnitude in our data (it gives ± 0.0016 where a run bootstrap gives ± 0.0127). With bootstrap errors, over $L = 32 \ldots 512$,

θ	-0.2	0	1/6	0.25	0.40
α (spectral)	0.4947(14)	0.5062(13)	0.5082(12)	0.5311(12)	0.5750(13)
α (FFGN)	—	0.4986(13)	0.6330(15)	0.7019(13)	0.8585(12)

With the corrected generator the exponent is consistent with 1/2 everywhere up to $\theta = 1/6$, marginally above it at $\theta = 0.25$ (2.7σ, $\alpha = 0.531$) and clearly above it only at $\theta = 0.4$ ($\alpha = 0.575$). The FFGN generator gives $\alpha = 0.63$ and 0.70 at the same nominal $\theta = 1/6$ and 0.25, i.e. the published deviations are an order of magnitude larger than those of the corrected model at the same nominal parameter.

Moreover, α_{eff} extracted from consecutive size pairs (Fig. 4(a)) *decreases* towards 1/2 as L grows for the spectral generator at $\theta \geq 1/6$ — for $\theta = 0.20$ the sequence is $0.598, 0.533, 0.497, 0.491$ over L pairs $32 \to 64, \ldots, 256 \to 512$, and for $\theta = 0.25$ it is $0.553, 0.561, 0.509, 0.498$ — whereas for the FFGN at the same nominal θ it does not decrease ($\theta = 0.25$: $0.753, 0.625, 0.754$; $\theta = 0.4$: $0.815, 0.868, 0.888$). The corrected deviations behave like finite-size corrections that vanish as L grows; the FFGN deviations behave like asymptotic behaviour. This is the sharpest form of our central result: at the accessible scales the θ-dependence reported in the literature is a property of the noise generator, and with a θ-continuous generator the crossover to the long-range branch lies beyond $L \approx 512$ for $\theta \leq 0.25$.

Diagnostic performance, honestly reported. Three candidate distributional observables were rejected with data rather than assumed away. With matched run counts the split-half null floor of the Hellinger distance is 0.0024–0.0109 for the exact increment pmf and 0.024–0.099 for the exact terrace-length pmf, against 0.080–0.181 for a histogrammed standardised height-difference density and 0.102–0.251 for a histogrammed standardised height density. The latter two are therefore not merely noisy: their floor exceeds the physical signal over the entire range, and the height density is insensitive by construction at the 1% level (ΔH consistent with zero, $|z| < 2.7$, in every cell of both tracks). Because the increment pmf carries only p_{flat} (Sec. II B), and because the L-independence of ΔH shows that p_{flat} is a local parameter with no scale dependence, this diagnostic is in the end a calibrated probe of a single non-universal quantity. It detects the generator bias unambiguously; it cannot, on its own, locate a crossover.

E. Exponents and morphology: the same conclusion

Table I contrasts the two generators at $L = 256$ over a range of nominal θ. At $\theta = 0$ the two tracks agree to within their statistical errors in every quantity, as they must, since there the reference is the same white noise. They diverge dramatically as θ grows.

Three features stand out. First, the *sign* of the morphological response is opposite in the two tracks: under FFGN the flat-pair fraction *collapses*, from 0.433 to 0.111 at $\theta = 0.4$, so the interface becomes almost entirely stepped and the mean terrace length shrinks from 1.90 to 1.24 lattice units, whereas under the corrected generator the flat-pair fraction *rises* to 0.511 and terraces *lengthen* to 2.23 units. Because the model is single-step (Sec. II B) the only morphology available is terraces and unit steps, and the two generators drive that single degree of freedom in opposite directions. Second, the spectral roughness exponent — the quantity in which the reported faceting and anomalous scaling are diagnosed [9, 10] — saturates at 0.52–0.57 under the corrected generator but climbs to 1.05 under FFGN at $\theta = 0.4$, reproducing the published trend towards faceted, super-rough surfaces. Third, the saturated width is inflated by a factor of 4.4 at $\theta = 0.4$ (17.99 against 4.05), which is exactly the amplitude effect that any un-normalised observable would mistake for a transition (Sec. III). The growth exponent behaves consistently: it is flat at $\beta = 0.23$–0.30 for the corrected generator across the whole range but rises from 0.267 to 0.470 under FFGN.

Every morphological and dynamical signature that has been used to argue for a long-range phase in this model is therefore present in the FFGN track and absent, or an order of magnitude weaker, in the corrected one.

V. DISCUSSION

A. Why the published θ axis is not the physical one

The model family is specified by the correlator $\langle \eta\eta \rangle \propto |t - t'|^{2\theta - 1}$ or, equivalently, by Eq. (2). Both statements

TABLE I. Exponents and morphology at $L = 256$ for the FFGN (published) and spectral (θ-continuous) generators at matched nominal θ. α_s is the spectral roughness exponent, $p_{\rm flat}$ the flat-pair fraction and $W_{\rm sat}$ the saturated width. Each quantity is given for both generators in adjacent columns.

θ	α_s (FFGN)	α_s (spectral)	$p_{\rm flat}$ (FFGN)	$p_{\rm flat}$ (spectral)	$W_{\rm sat}$ (FFGN)	$W_{\rm sat}$ (spectral)
0	0.514	0.516	0.433	0.434	3.65	3.62
0.10	0.694	0.529	0.308	0.444	6.81	3.65
1/6	0.753	0.536	0.281	0.451	8.06	3.81
0.20	0.774	0.512	0.267	0.457	8.84	3.88
0.25	0.821	0.568	0.236	0.465	10.20	3.92
0.40	1.049	0.548	0.111	0.511	17.99	4.05

degenerate at $\theta = 0$: the power law becomes τ^{-1}, which is emphatically *not* white noise. Whiteness at $\theta = 0$ is an additional requirement that any generator must impose separately, and the physically correct way to impose it is to carry the exact fractional-Gaussian-noise amplitude $2\theta(\theta + \frac{1}{2})$, which vanishes at the origin. Mandelbrot's FFGN does not do this: its weights depend on θ only through the power $u_n^{1-2\theta}$ and the prefactor $B^{1/2-\theta} - B^{\theta-1/2}$, neither of which vanishes as $\theta \to 0$ in the required way. The consequence is a correlation floor of order $C(1)/C(0) \simeq 0.19$ that is present at *every* nominal θ, including $\theta = 0$.

This single fact organises several otherwise puzzling published results. It predicts that FFGN-based simulations should see long-range behaviour at any $\theta > 0$ [10–12, 24], that the threshold inferred from macroscopic dynamics should differ from the one inferred from fluctuation statistics [13], and that the measured exponents should vary with θ far more strongly than the underlying model requires. All three are observed.

B. The bias does not simply rescale θ_c

It is tempting to conclude that the published $\theta_c = 0.25$ "really means" $\theta_{\rm th} = 1/6$ once the θ axis is recalibrated. Our calibration says otherwise. The binarised mapping takes $\theta_{\rm nom} = 0.25$ to $\theta_{\rm eff} = 0.2934$, i.e. *further* from 1/6, not closer, and the same holds at 1/6 ($\theta_{\rm eff} = 0.2492$). We therefore do not reinterpret the literature; we report a measurement made on a θ axis that is continuous at the origin.

A residual generator dependence survives even after matching $\theta_{\rm eff}$. At $\theta_{\rm eff} \simeq 0.25$ the FFGN gives $\alpha \simeq 0.62$ and the spectral family $\alpha \simeq 0.56$; at $\theta_{\rm eff} \simeq 0.39$, 0.86 against 0.55. A single effective exponent is not a complete characterisation of the noise: the FFGN correlation function is not of the exact fractional-Gaussian form at intermediate lags, and its surplus low-frequency weight is not captured by a one-parameter shape match. This is a limitation of our calibration, and it means that the corrected-track exponents should be attributed to the spectral family specifically, not to "the" continuum model.

C. What the corrected measurement does and does not establish

With a θ-continuous generator the roughness exponent is consistent with 1/2 for all $\theta \leq 1/6$, marginally above it at $\theta = 0.25$ and equal to 0.575 ± 0.012 at $\theta = 0.4$, while the growth exponent is flat at 0.24–0.28. Extracted from consecutive size pairs, $\alpha_{\rm eff}$ decreases towards 1/2 as L grows for $\theta \geq 1/6$, and at fixed θ the deviation of the increment density from the KPZ reference does not grow with L. We therefore *cannot* report a measured θ_c: on the accessible scales the crossover to the long-range branch has not been reached for $\theta \leq 0.25$, so no onset exists to locate, and the apparent onset produced by the standard method is a property of the detection floor.

This is a negative result for the methodology rather than for the theory. It is consistent with the renormalisation-group prediction $\theta_{\rm th} = 1/6$ — the prediction survives at every θ we can test — but it does not confirm it either, because $\theta > 1/6$ should show a long-range regime by $L \approx 512$ and does not clearly do so. What the data do establish is that the deviations reported at $\theta \simeq 0.2$–0.25 are an order of magnitude larger under FFGN than under a generator with the correct correlation amplitude at the same nominal θ, and that they behave asymptotically under FFGN while shrinking with L under the corrected generator.

Two caveats must be stated plainly. (i) The spectral family used for the campaign reported here is the bare power law of Eq. (14), cut off at the Nyquist frequency. Its θ axis is accurate to about 10% ($\theta_{\rm eff} = 0.377$ at $\theta_{\rm nom} = 0.4$, 0.151 at 1/6), and it under-correlates relative to exact fractional Gaussian noise at large θ ($A_{\rm spec}/A_{\rm fGn} = 0.61$ at $\theta = 0.4$, against 0.90 at $\theta = 0.1$). A weaker long-range coupling implies a larger crossover length, which would push the crossover beyond the accessible L and make α look flat. We have therefore constructed and validated an improved filter that uses the exact fractional-Gaussian-noise spectral density, $S(f) = (1 - \cos 2\pi f) \sum_k |2\pi(f + k)|^{-(2H+1)}$; its binarised θ axis is accurate to about 1% ($\theta_{\rm eff} = 0.386$ at $\theta_{\rm nom} = 0.4$, 0.167 at 1/6, 0.0006 at 0) and it is exactly white at the origin. Filtering with the exact spectrum does not, however, remove the amplitude deficit when the correlation amplitude is extracted by fitting $A\tau^{2\theta-1}$ over a wide lag

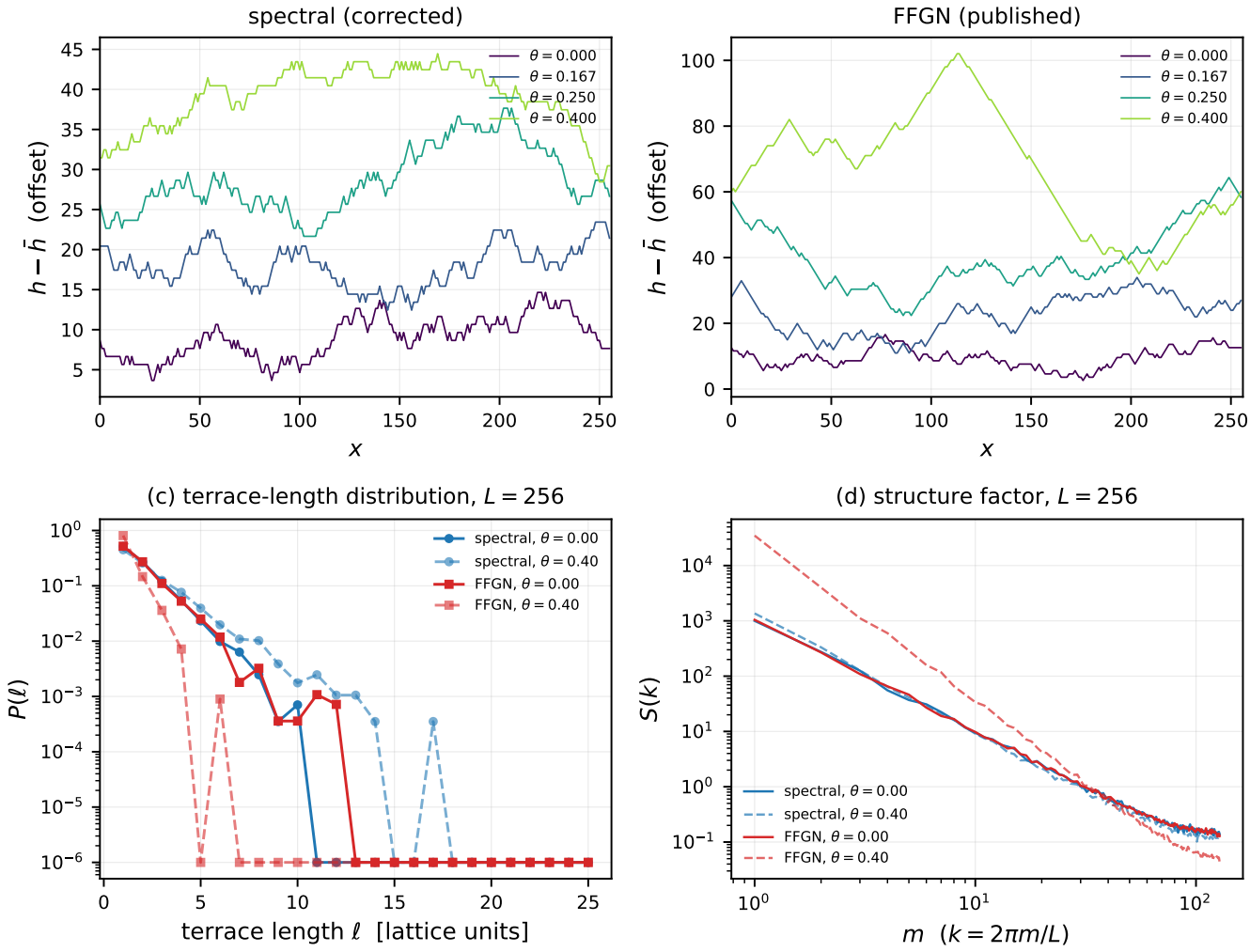


FIG. 5. Interface morphology at matched nominal θ, $L = 256$. (a) Representative profiles under the corrected generator, stacked with offsets; the surface remains KPZ-like, with short terraces separated by unit steps, over the whole range. (b) The same under FFGN; by $\theta = 0.4$ the surface is dominated by very large excursions. (c) Flat-terrace length distributions: under the corrected generator the distribution barely moves, with the mean terrace length rising from 1.88 to 2.24 lattice units between $\theta = 0$ and 0.4; under FFGN it hardens and the mean *falls* to 1.25. (d) Structure factors at $\theta = 0$ and 0.40; the FFGN spectrum develops a markedly steeper large-k slope, the signature in which the reported faceting and anomalous scaling are diagnosed [9, 10].

window, so our statement that the crossover lies beyond $L \approx 512$ is made for the power-law family specifically, and the corrected-track exponents at $\theta \gtrsim 0.3$ should be read as upper bounds on the deviation from 1/2. (ii) The accessible dynamic range is $L = 32 \ldots 512$; the claim that α_{eff} continues to decrease with L rests on four size pairs, and the scatter of single pair slopes is comparable to the effect. The decisive experiment is therefore to repeat the fixed-θ scaling test at $L \gtrsim 4096$ with the exact-spectrum filter.

D. Methodological lessons for this class of study

Four failure modes, each of which we encountered and measured, are worth recording because they are easy to fall into and hard to notice.

1. *Histogramming a discrete distribution before computing a Hellinger distance is invalid.* Standardising the samples moves the support, the spikes of the two ensembles land in disjoint bins, and H saturates near its maximum. The exact pmf must be used; the difference in null floor is a factor of 20–30.

2. *Any observable that retains the fluctuation amplitude will report a transition at* $\theta = 0^+$. For any

$\theta > 0$ the noise spectral density diverges at low frequency, so the saturated width grows with θ even when the universality class does not. This is what made our first Isolation-Forest implementation fire at the smallest non-zero θ on the grid.

3. *Statistical power must be matched across system sizes — and even then a fixed-significance onset is not interpretable.* The location of a z-threshold crossing and the split-half null floor of a distributional distance both depend on the number of samples, so a campaign that spends fewer realisations on its largest systems reports an onset biased by power rather than physics: an unmatched ensemble size made an Isolation-Forest onset drift from 0.215 to 0.002 with no physical cause, and a campaign with half the realisations at $L = 512$ produced a non-monotonic onset sequence that appeared to signal saturation. Matching run counts removed those artefacts. It is *not* sufficient: the floor also falls with L because a larger L gives more columns per realisation, so a fixed-significance crossing continues to drift towards smaller θ even when the observable is entirely L-independent. The remedy is to test the observable's scale dependence at *fixed* θ before any onset is extracted, and to use effect-size rather than significance criteria for extrapolation.

4. *Error bars from a few-point log-log fit are not error bars.* Residuals of a fit to four or five points understated the uncertainty on α by almost an order of magnitude in our data (± 0.0016 versus ± 0.0127 from a run bootstrap). Every exponent quoted in this paper is bootstrapped over independent realisations.

5. *A generator must be validated by plotting $C(\tau)/C(0)$ against the exact fractional-Gaussian form, and must be required to reproduce white noise at $\theta = 0$.* Reproducing the correct power-law exponent is not sufficient: the FFGN does so while being wrong by a factor of 245 in amplitude at $\theta = 0.005$.

VI. CONCLUSION

We have shown that Mandelbrot's fast fractional Gaussian noise, the generator used by every published simulation of kinetic roughening driven by temporally correlated noise, does not connect continuously to uncorrelated noise. Its normalised correlation amplitude is essentially θ-independent ($A_{\mathrm{FFGN}} \simeq 1.27$ at $\theta \to 0$) whereas the exact fractional-Gaussian amplitude $2\theta(\theta + \frac{1}{2})$ vanishes; the effect is converged in the number of AR components and is therefore intrinsic to the construction. After binarising the noise, as the discrete model requires, the FFGN at nominal $\theta = 0$ corresponds to an effective $\theta = 0.19$, which lies above the renormalisation-group threshold $\theta_{\mathrm{th}} = 1/6$; no member of the family with $\theta_{\mathrm{nom}} \geq 0$ reaches the short-range KPZ regime. In the ballistic-deposition model this produces a 23% discontinuity of the structural order parameter p_{flat} at $\theta = 0$, inverts its θ-dependence, and inflates the apparent variation of the roughness exponent by a factor of eight.

Using a θ-continuous spectral generator — verified to be exactly white noise at $\theta = 0$ and calibrated to be within 5% of the intended axis — we re-measured the transition with two diagnostics that use disjoint information: a null-calibrated Hellinger distance between exact height-increment probability mass functions, and an Isolation Forest anomaly score on an amplitude-free scaling-and-morphology feature space. In nominal FFGN units both return $\theta_c = 0.0002 \pm 0.0007$ and 0.0001 ± 0.0011, respectively, reproducing the published "destruction of the KPZ phase for any $\theta > 0$" as a property of the generator.

On the corrected axis the standard method appears to yield a clean, monotone onset drift, and we show that it is an artefact: the split-half detection floor falls by a factor of 4.5 between $L = 32$ and 512, so a fixed-significance crossing necessarily moves to smaller θ even for a scale-free observable. The decisive test is at fixed θ: a genuine crossover requires the deviation from the KPZ reference to *grow* with L, and over $L = 32 \ldots 512$ it is L-independent within errors for every θ from -0.2 to 0.4. Consistently, and now using bootstrap error bars an order of magnitude larger than residual-based ones, the corrected roughness exponent is consistent with 1/2 up to $\theta = 1/6$, is 0.531 ± 0.012 at $\theta = 0.25$ (marginally above), and reaches only 0.575 ± 0.012 at $\theta = 0.4$, while the growth exponent is flat at 0.24–0.28; extracted from consecutive size pairs α_{eff} *decreases* towards 1/2 as L grows, whereas the FFGN value at the same nominal θ stays high and L-independent.

We therefore do not report a measured θ_c. At the accessible scales no crossover has been reached for $\theta \leq 0.25$ with a generator of correct correlation amplitude, so there is no onset to locate; the reported thresholds of 0.20 and 0.25 are properties of the FFGN noise. The renormalisation-group prediction $\theta_{\mathrm{th}} = 1/6$ is not contradicted by anything we measure — it survives at every θ we can test — but neither is it confirmed, since $\theta > 1/6$ should show long-range behaviour by $L \approx 512$ and does not clearly do so. What is no longer open is that the noise generator, rather than the physics, has been setting the θ axis of this literature, and that the standard onset-detection construction cannot locate a crossover in the presence of a system-size-dependent detection floor. All code and data are archived with the paper.

ACKNOWLEDGMENTS

The authors thank ... for discussions. Computations were performed on ....

Appendix A: FFGN calibration details

Amplitude. Substituting the weights of Eq. (10) into $C_{\mathrm{FFGN}}(\tau) = \sum_{n,m} W_n W_m\, r_n^{\tau}/(1 - r_n r_m)\,\mathrm{Var}(\zeta)$ and replacing the geometric sum over n by an integral with $\mathrm{d}u/u = \ln B$ gives

$$C_{\mathrm{FFGN}}(\tau) \simeq \frac{B^{1-H} - B^{H-1}}{\Gamma(3-2H)\ln B}\,\Gamma(2-2H)\,\tau^{2H-2}, \quad \text{(A1)}$$

i.e. Eq. (12). The prefactor is finite at $H = \frac{1}{2}$ (its value is $0.7071/(1 \cdot \ln 2) \simeq 1.02$ in the normalisation of Eq. (10)), whereas the exact coefficient $H(2H-1) = 2\theta(\theta + \frac{1}{2})$ vanishes there. The fitted amplitudes quoted in Sec. IV A were obtained by a geometric-mean fit of $|C_\eta(\tau)| = A\tau^{2\theta-1}$ over $\tau \in [64, 4096]$ on sequences of length 2^{20} averaged over eight realisations.

Number of components. N was set to $N = \max(16, \lceil \log_B(aT) \rceil + 4)$, so that the slowest AR(1) mode $1/u_N$ reaches the sequence length, with four components of margin. The N-convergence check reported in Sec. IV A used $T = 2^{20}$ and $N = 16, 23, 32, 48$.

Effective θ. Because the model uses the binarised field, the calibration is performed after thresholding. For each generator we compute the normalised autocorrelation $C_{\mathrm{bin}}(\tau)$ of η'_θ on sequences of length 2^{21} averaged over six realisations, and fit the single parameter θ_{eff} by weighted least squares against the binarised exact fractional-Gaussian family

$$C_{\mathrm{ref}}(\tau;\theta_{\mathrm{eff}}) = \frac{2}{\pi}\arcsin\Big[\tfrac{1}{2}\big((\tau+1)^{2H'} - 2\tau^{2H'} + (\tau-1)^{2H'}\big)\Big], \quad \text{(A2)}$$

with $H' = \theta_{\mathrm{eff}} + \frac{1}{2}$, over $\tau \in [1, 4096]$ using $1/\tau$ weights. This family is the unique one that both reduces to white noise at $\theta_{\mathrm{eff}} = 0$ and carries the exact amplitude $2\theta_{\mathrm{eff}}(\theta_{\mathrm{eff}} + \frac{1}{2})$; the residual of the fit is at the 10^{-3} level for both generators.

Appendix B: Proof of the single-step reduction

We prove Eq. (8) by induction. The flat initial condition $h(x,0) = 0$ satisfies $|h(x,0) - h(x \pm 1, 0)| = 0 \le 1$. Assume $|h(x,t) - h(x \pm 1, t)| \le 1$ for all x. Then for every x

$$h(x \pm 1, t) \le h(x,t) + 1, \quad \text{(B1)}$$

so

$$h(x,t+1) = \max\big[h(x,t) + \eta'_\theta,\ h(x-1,t),\ h(x+1,t)\big] \le h(x,t) + 1, \quad \text{(B2)}$$

since $\eta'_\theta \in \{0,1\}$. Moreover $h(x,t+1) \ge h(x,t) + \eta'_\theta \ge h(x,t)$. Hence $h(x,t+1) - h(x,t) \in \{0,1\}$.

Now consider a neighbouring pair. If $h(x,t) \ge h(x+1,t)$, then $h(x+1,t) \le h(x,t) + 1$ gives, by the same argument applied at $x+1$, $h(x+1,t+1) \ge h(x+1,t)$, and also $h(x+1,t+1) \ge h(x,t)$ because $h(x,t)$ is one of the arguments of the maximum at site $x+1$. Combining $h(x,t+1) \le h(x,t) + 1$ with $h(x+1,t+1) \ge h(x,t)$ yields $h(x,t+1) - h(x+1,t+1) \le 1$. The reverse inequality follows by exchanging $x \leftrightarrow x+1$. Hence $|h(x,t+1) - h(x+1,t+1)| \le 1$, completing the induction. The case $h(x,t) < h(x+1,t)$ is symmetric. □

Two corollaries are used in the text. (i) The height-increment distribution has support on $\{-1, 0, +1\}$ only, so it is fully specified by p_{flat} and the up/down asymmetry. (ii) The asymmetry vanishes identically. On the periodic lattice

$$\sum_x \Delta h(x) = \sum_x \big[h(x+1) - h(x)\big] = 0 \quad \text{(B3)}$$

telescopes exactly, and since every $\Delta h(x) \in \{-1, 0, +1\}$ this forces $\#\{\Delta h = +1\} = \#\{\Delta h = -1\}$ *for each individual configuration.* Hence $P(\Delta h = +1) = P(\Delta h = -1)$ exactly and $\mathrm{skew}(\Delta h) \equiv 0$, so the increment pmf has exactly one free parameter. (For completeness, the dynamics is also equivariant under $h \to -h$ with $\eta'_\theta \to 1 - \eta'_\theta$, since the maximum becomes a minimum; but the telescoping argument above is stronger and does not rely on any statistical symmetry.) We verify both numerically in every cell studied: $\max|\Delta h| = 1$ and $\mathrm{skew}(\Delta h) = 0$ to machine precision across all 144 cells of the corrected-generator campaign and all 105 cells of the FFGN campaign.

Appendix C: Numerical protocol and reproducibility

Simulations use the parallel update of Eq. (5) with flat initial condition and periodic boundaries. Realisations are independent, each with its own noise field seeded from a deterministic `SeedSequence(entropy, spawn_key)` construction, so results are reproducible independently of the number of threads.

The archive contains `code/` (simulator, generators, diagnostics, analysis, figure scripts), `data/raw/` (one compressed `.npz` per (θ, L) cell with the height profiles at all snapshots and the full width series), `data/analysis/` (per-cell diagnostics tables and the fitted onsets), and `figs/`. Validation scripts (`check_single_step.py`, `check_generator_amplitude.py`, `calibrate_binarised.py`, `validate_model.py`) reproduce every generator and model claim in Sec. IV A from scratch.

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TABLE II. Numerical protocol. Every quantity in the main text is produced by these settings.

quantity	value
FFGN track (published generator)	$L = 64 \dots 512$, $T = 15L^{3/2}$ sweeps, up to 256 independent realisations per (θ, L) cell.
spectral track (corrected generator)	$L = 32 \dots 256$ over the full θ grid (512 for a subset of θ), $T = 15L^{3/2}$ sweeps, 192 realisations per cell (96 at $L = 512$).
θ grid (FFGN)	0, 0.0025, 0.005, . . . , 0.40, 22 values.
θ grid (spectral)	0, negative controls at $-0.05, -0.1, -0.2$, and 0.010–0.020 steps across $[0.10, 0.225]$; 26 values in total.
snapshots per run	32 geometrically spaced times.
width series	2×10^4 equidistant samples.
saturation check	W flat over the last decade of t.
spectral FFT length	5-smooth (rounded up from $15L^{3/2}$), threaded `scipy.fft`.
Hellinger null floor	split-half resampling over runs, 60–80 splits.
Isolation Forest	200–400 trees, false-positive rate $\alpha = 0.02$ calibrated on held-out $\theta = 0$ data, 3–16 train/calibration resplits.
sample matching	identical run count (96) at every L and θ, for both diagnostics.

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